ForensiQ

DIGITAL FORENSIC INVESTIGATION TOOLKIT

- ACQUIRE
- ANALYZE
- CORRELATE
- REPORT

From Evidence to Answers

1. INTRODUCTION

Digital Forensic Investigation Toolkit — ForensiQ

An integrated tool for automated evidence acquisition, multi-source analysis, artifact correlation and court-ready reporting.

ACQUIRE.
ANALYZE.
CORRELATE.
REPORT.

What is ForensiQ?

ForensiQ is a digital forensic toolkit that helps investigators collect, analyze and correlate digital evidence from disk, memory and network sources. It provides an easy-to-use interface to perform forensic analysis and generate structured, court-ready reports.

Automates complex tasks
Reduces manual effort and errors

Maintains evidence integrity
Hash verification & chain-of-custody logs

Unified analysis
Disk, memory and network artifacts in one platform

Generates legal reports
Structured, court-ready PDF reports

How to Use ForensiQ?

1. **Create a Case**
   Enter case details and select evidence type (disk / memory / network).
2. **Acquire Evidence**
   Collect data through write-blocked mode and verify hash (SHA-256 / SHA-1).
3. **Run Analysis**
   Parse files, analyze memory dumps, inspect network captures and recover artifacts.
4. **View Results**
   See timelines, artifacts, IOC matches and detailed analysis results in the dashboard.
5. **Generate Report**
   Create a structured, court-ready PDF report with evidence details and chain-of-custody records.

ForensiQ

Digital Forensic Investigation Toolkit

Acquire. Analyze. Correlate. Report.

2. PROJECT OVERVIEW & USAGE

ForensiQ is a unified digital forensic toolkit that automates evidence acquisition, multi-source analysis, artifact correlation, and generation of court-ready reports.

FASTER INVESTIGATIONS.
STRONGER EVIDENCE.
CLEARER INSIGHTS.
COURT-READY REPORTS.

What is ForensiQ?

ForensiQ is an integrated digital forensic investigation platform that helps investigators acquire, analyze and correlate evidence from disk, memory and network sources through a simple, user-friendly interface.

Key Features

- Write-blocked evidence acquisition with hash verification
- Tamper-evident chain-of-custody logging
- Unified disk, memory and network analysis
- Cross-source timeline reconstruction
- IOC & keyword correlation
- Structured, court-ready PDF report generation

Who Can Use?

- Law enforcement investigators
- Corporate security & incident response teams
- Academic and research labs
- Digital forensics trainees and professionals

How to Use ForensiQ? (Step-by-Step Workflow)

1 Connect Device

Connect your Android device using USB to start investigation.



Shows device connection status and guides you to connect the Android device via USB.

2 Device Overview

View basic information about the connected device.



Shows device model, Android version, security patch, boot status, SELinux status and more.

3 Collect Evidence

Collect installed apps, files, logs and other artifacts.



Click "Collect from Android" to start acquiring application and system data.

4 Analyze & Investigate

Analyze collected data for security and forensic insights.



View device security posture and analyze apps, permissions, processes, network activity and more.

5 Generate Report

Create a structured, court-ready forensic report.



Compile analysis results into a structured PDF report with evidence details and chain-of-custody records.

System Workflow (End-to-End)

ForensiQ

Digital Forensic Investigation Toolkit

Acquire. Analyze. Correlate. Report.

03

SYSTEM METHODOLOGY

From Evidence Acquisition to Forensic Intelligence

A structured workflow that automates evidence acquisition, multi-source analysis, artifact correlation and generation of court-ready reports.

EVIDENCE
ANALYSIS
CORRELATION
INTELLIGENCE

1 Evidence Acquisition

- Acquire disk, memory and network evidence.
- Use write-blocked acquisition methods.



2 Evidence Integrity Verification

- Calculate SHA-256 / SHA-1 hashes.
- Verify evidence integrity.



3 Chain-of-Custody Logging

- Record acquisition timestamps.
- Maintain investigator and evidence handling records.
- Preserve an auditable evidence trail.



4 Artifact Extraction & Analysis

- Parse NTFS, EXT and FAT filesystems.
- Recover deleted files using file carving.
- Analyze memory dumps (Volatility-based pipeline).
- Examine network PCAP captures.



5 Timeline Reconstruction

- Correlate filesystem, registry and log timestamps.
- Build a unified investigative timeline.



6 IOC & Artifact Correlation

- Search suspicious indicators and keywords.
- Identify relationships between extracted artifacts.



7 Forensic Report Generation

- Generate structured PDF reports.
- Include evidence details, analysis findings and chain-of-custody records.



Investigation Workflow (End-to-End)



ACQUIRE
Disk / Memory / Network

VERIFY
Hash Verification

ANALYZE
Artifacts & Evidence

CORRELATE
Timeline & iOC Matching

REPORT
Court-Ready PDF

ForensiQ

Digital Forensic Investigation Toolkit

Acquire. Analyze. Correlate. Report.

04 IMPLEMENTATION & PROJECT OUTCOMES

Turning Digital Evidence into Actionable Intelligence

EVIDENCE.
ANALYSIS.
CORRELATION.
INTELLIGENCE.

A. CORE IMPLEMENTATION

Multi-Source Forensic Analysis Engine

Disk Forensics

- Filesystem parsing (NTFS / EXT / FAT)
- Deleted-file recovery
- File carving

Disk Images

Memory Forensics

- RAM analysis
- Process inspection
- Network connections
- Suspicious artifacts detection

Memory Dumps

Network Forensics

- PCAP parsing
- Protocol analysis
- Session reconstruction
- Suspicious communication detection

PCAP Captures

Timeline & Correlation

- Cross-source event reconstruction
- Correlate filesystem, registry and logs
- IOC & keyword matching

Unified Timeline

B. SECURITY & EVIDENCE INTEGRITY

Trusted. Tamper-Evident. Secure.

SHA-256 / SHA-1

- Cryptographic hashing for evidence verification
- Ensures data integrity

Chain-of-Custody Logging

- Tamper-evident records
- Acquisition timestamps
- Auditable evidence trail

JWT Authentication

- Secure investigator login
- Role-based access control
- API authentication

S3-Compatible Storage

- Secure evidence storage
- Large file handling (disk images, PCAP, etc.)

Evidence Traceability & Audit Records

- Track every action performed on evidence
- Maintain complete investigation logs



C. EXPECTED RESULTS

A More Efficient and Reliable Forensic Process

Automated Evidence Acquisition
Reduces manual effort and errors

Reliable Evidence Integrity
Hash verification & chain-of-custody logs

Unified Multi-Source Analysis
Disk, memory and network artifacts in one platform

Faster Investigation Workflow
Automates repetitive tasks

Structured PDF Reporting
Court-ready reports with evidence details

Reduced Manual Workload
Simplifies analysis and documentation

D. PROJECT IMPACT

From Evidence to Actionable Intelligence

- Reduces repetitive manual forensic tasks.
- Combines multiple forensic analysis activities into one platform.
- Improves evidence tracking and documentation.
- Supports consistent and legally admissible investigation reporting.
- Helps investigators understand complex digital evidence.

FASTER INVESTIGATIONS.
STRONGER EVIDENCE.
CLEARER INSIGHTS.
COURT-READY REPORTS.

ACQUIRE → **VERIFY** → **ANALYZE** → **CORRELATE** → **REPORT**

Disk / Memory / Network | Hash Verification | Artifacts & Evidence | Timeline & IOC Matching | Court-Ready PDF

ForensiQ – Technology Stack

Digital Forensic Investigation Toolkit with Automated Evidence Acquisition, Multi-Source Analysis & Court-Ready Reporting

Acquire. Analyze. Correlate. Report.

Frontend (User Interface)

Flutter (Dart)

Cross-platform dashboard for investigators

- Android App
- Windows App
- Web (Flutter Web)

- Case management UI
- Evidence acquisition controls
- Analysis reports & visualization
- PDF report viewer
- Multi-platform support

HTTP/JSON (REST API)

Backend (API & Core Services)

Rust (Axum)

High-performance REST API and forensic processing engine

- REST API
- Authentication (JWT)
- Case Management
- Evidence Processing
- Artifact Analysis
- Report Generation

SQLx (Async)

Database

PostgreSQL

Stores case data, evidence metadata, and chain-of-custody logs

- Case details & investigation data
- Evidence metadata
- Chain-of-custody records
- User accounts & permissions
- Analysis results & reports

Forensic Processing Modules

Disk Forensics

- The Sleuth Kit (libtsk)
- Filesystem parsing
- Deleted file recovery
- File carving

Memory Forensics

- Volatility Framework
- Process analysis
- Network connections
- Injected artifacts

Network Forensics

- PCAP analysis
- Protocol parsing
- Session reconstruction
- Suspicious traffic detection

Timeline & Correlation

- Cross-source timeline
- IOC & keyword search
- Correlation engine
- Artifact relationship mapping

Authentication & Storage

JWT

- Secure login & role-based access
- API authentication

Amazon S3 (Compatible)

- Evidence file storage
- Large file handling (disk images, PCAP, etc.)

External Forensic Tools & Libraries

The Sleuth Kit (libtsk)
Filesystem analysis & file carving

Volatility Framework
Memory dump analysis

Wireshark / TShark
PCAP parsing & network analysis

Python Scripts
Supporting scripts and plugins

Reporting & Visualization

PDF Report Generation

- Structured, court-ready reports
- Evidence summary
- Timeline, artifacts, IOC results
- Auto-included chain-of-custody

Charts & Visualizations

- Timeline view
- Artifact correlations
- Interactive dashboards
- Case statistics

ForensiQ

Digital Forensic Investigation Toolkit

with Automated Evidence Acquisition, Multi-Source Analysis, Artifact Correlation & Court-Ready Reporting

ACQUIRE • ANALYZE • CORRELATE • REPORT

Key Features

01 **Automated Evidence Acquisition**
Write-blocked collection with SHA-256/SHA-1 verification

02 **Chain-of-Custody Logging**
Tamper-evident, auto-generated, auditable logs

03 **Multi-Source Analysis**
Disk, memory and network artifact analysis in a unified workflow

04 **Artifact Extraction & File Recovery**
Filesystem parsing and file carving (NTFS/EXT/FAT)

05 **Memory & Network Forensics**
Process analysis, network connections, PCAP analysis

06 **Timeline Reconstruction**
Cross-source correlation of filesystem, registry and logs

07 **IOC & Keyword Correlation**
Automatic detection and correlation across all artifacts

08 **Court-Ready Reporting**
Structured PDF reports with verified evidence and chain-of-custody

Designed by Team Members

G. SAI	**Team Lead** & Backend Development (Rust, API, Database, Integration)
K. SHANMUKH RAO	**Frontend** Development (Flutter UI, Dashboard)
K. CHARAN TEJA	**Forensic Modules** (Disk, Memory & Network Analysis)
G. SHARAN KUMAR	**Security & Authentication** (JWT, Hashing, Chain-of-Custody)
J. DINESH	**Report Generation** & Testing (PDF Reports, Documentation, QA)

Thank You

FOR YOUR TIME AND SUPPORT